

Programs 8, 9 and 10

Visual Basic 2010 front end with Oracle 10g Express Edition back end

Step-by-step explanation

Program	Topic	Oracle table
8	Library: add, issue, return, view books	book
9	Student marks: total, result, percentage	st
10	Employee payroll: allowance, deduction, gross, net	emp

Contents: Common setup (Oracle + VB 2010) / Program 8 / Program 9 / Program 10

Common setup (needed for all three programs)

1. Software needed

1. Oracle Database 10g Express Edition (XE) installed. During installation you set the password for the **SYSTEM** user. The code uses password **system**, so either set that, or change it in the connection string.
2. Visual Basic 2010 (Visual Studio 2010 / VB 2010 Express).
3. Oracle services running: **OracleServiceXE** (the database) and **OracleXETNSListener** (the listener). Check in Start > Run > services.msc. If they are stopped, VB will show ORA-12541 or ORA-12154.

2. Create the table in Oracle 10g XE

1. Start > All Programs > Oracle Database 10g Express Edition > **Run SQL Command Line**.
2. Log in: SQL> connect system/system (use your own password if different).
3. Paste the CREATE TABLE statement of the program and press Enter. Oracle replies **Table created**.
4. Check the structure with SQL> desc book; (Or st / emp).
5. Later, after running the VB form, check the stored data with SQL> select * from book;.

Alternative: open the browser page *Get Started with Oracle Database 10g Express Edition* > Database Home Page > SQL > SQL Commands, paste the statement and click Run.

3. Create the VB 2010 project

1. File > New Project > **Windows Forms Application**. Keep the name **Form1**.
2. Drag the Labels, TextBoxes, Buttons and DataGridView from the Toolbox onto the form (design given in each program). **Do not rename the controls**. Keep the default names TextBox1, TextBox2 ... Button1 ... because the code uses these names.
3. Double-click each Button. VB creates an empty Button1_Click event. Paste the code of that button inside it.
4. Put the line Imports System.Data.OleDb as the very first line of the code window (above Public Class Form1) and the Dim cn ... line just below Public Class Form1.
5. Press **F5** to run.

4. Parts of the code that are the same in every program

<pre>Imports System.Data.OleDb Public Class Form1 Dim cn As New OleDbConnection("Provider=MSDAORA;Data Source=XE;User ID=system;Password=system;")</pre>	
Part	Meaning
Imports System.Data.OleDb	Brings in the OLE DB classes (OleDbConnection, OleDbCommand, OleDbDataAdapter). Without it you would have to write System.Data.OleDb. before every class.
Dim cn As New OleDbConnection(...)	Creates the connection object cn at form level, so every button can use it. It is only created here; the database is opened later with cn.Open().
Provider=MSDAORA	Microsoft OLE DB Provider for Oracle. It tells VB which driver to use to talk to Oracle.
Data Source=XE	The Oracle service name. XE is the default name of Oracle 10g Express.
User ID / Password	The Oracle login. Here the SYSTEM user is used, so the table is created in the SYSTEM schema.

5. The five objects used in the programs

Object / method	What it does
OleDbConnection	The link between VB and Oracle. Open () starts it and Close () ends it.
OleDbCommand(sql, cn)	Holds one SQL statement and the connection it should run on.
ExecuteNonQuery()	Runs INSERT, UPDATE or DELETE. It returns the number of rows affected (used to test 'record found or not').
OleDbDataAdapter + DataTable	Used for SELECT. da.Fill(dt) runs the query and puts the rows in the DataTable. The adapter opens and closes the connection by itself.
DataGridView1.DataSource = dt	Shows the DataTable as a grid on the form.
Try / Catch / Finally	Try runs the database code. If Oracle gives an error, Catch shows the message instead of crashing. Finally always runs, so the connection is closed in success and in failure.

Important note 1 (Provider not found / not registered). MSDAORA is a 32-bit provider. If VB shows '*The 'MSDAORA' provider is not registered on the local machine*' on a 64-bit Windows, set Project > Properties > Compile > Advanced Compile Options > **Target CPU = x86** and run again.

Important note 2. These three programs send ordinary SQL statements (INSERT, UPDATE, DELETE, SELECT) from VB to Oracle. They do not call a PL/SQL *stored procedure*. If your lab sheet asks specifically for a stored procedure with CREATE PROCEDURE, tell me and I will make that version.

Program 8: Library Management (Book table)

Aim: To create a VB 2010 form connected to Oracle 10g XE that adds a book, issues a book (quantity decreases by 1), returns a book (quantity increases by 1) and shows a report of all books.

Step 1: Create the table in Oracle

```
create table book (  
    bookid    number primary key,  
    bookname  varchar2(30),  
    author    varchar2(20),  
    quan      number  
);
```

Column	Type	Meaning
bookid	NUMBER, primary key	Unique book number. Oracle rejects a duplicate id (ORA-00001).
bookname	VARCHAR2(30)	Book title, maximum 30 characters.
author	VARCHAR2(20)	Author name, maximum 20 characters.
quan	NUMBER	Number of copies available. Issue reduces it, return increases it.

Step 2: Design the form

Control	Name	Text property (caption)
Form	Form1	Library Management
TextBox	TextBox1	Book ID
TextBox	TextBox2	Book Name
TextBox	TextBox3	Author
TextBox	TextBox4	Quantity
Button	Button1	Add Book
Button	Button2	Issue Book
Button	Button3	Return Book
Button	Button4	View Report
DataGridView	DataGridView1	Placed at the bottom of the form to show the records

Place one Label beside each TextBox (Book ID, Book Name, Author, Quantity) so the user knows what to type.

Step 3: Code, connection

```
Imports System.Data.OleDb  
  
Public Class Form1  
    Dim cn As New OleDbConnection("Provider=MSDAORA;Data Source=XE;User ID=system;Password=system;")
```

Explained in the common setup, section 4.

Step 4: Button1 - Add Book

```
Private Sub Button1_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button1.Click
    If TextBox1.Text = "" Or TextBox2.Text = "" Or TextBox3.Text = "" Or TextBox4.Text = "" Then
        MsgBox("Please fill in all fields before adding a book.")
        Exit Sub
    End If

    Try
        cn.Open()
        Dim sql As String = "INSERT INTO book VALUES (" & TextBox1.Text & ", '" & TextBox2.Text & "', '"
            & TextBox3.Text & "', " & TextBox4.Text & ")"
        Dim cmd As New OleDbCommand(sql, cn)
        cmd.ExecuteNonQuery()

        MsgBox("Book Added Successfully!")

        TextBox1.Clear()
        TextBox2.Clear()
        TextBox3.Clear()
        TextBox4.Clear()
    Catch ex As Exception
        MsgBox("Error adding book: " & ex.Message)
    Finally
        cn.Close()
    End Try
End Sub
```

1. Handles Button1.Click links this procedure to the Add Book button. It runs when the button is clicked.
2. The first If checks whether any of the four textboxes is empty. If yes, a message is shown and Exit Sub stops the code. This matters because an empty box makes the SQL invalid, for example VALUES (101, 'DBMS', , 5), and Oracle gives **ORA-00936: missing expression**.
3. cn.Open() opens the connection to Oracle.
4. The sql string is built by joining text with &. Number columns (bookid, quan) are written without quotes. Text columns (bookname, author) are inside single quotes. For input 101, DBMS Concepts, Korth, 5 the final string is: INSERT INTO book VALUES (101, 'DBMS Concepts', 'Korth', 5)
5. New OleDbCommand(sql, cn) prepares that statement on the open connection.
6. cmd.ExecuteNonQuery() sends it to Oracle and the row is inserted (OLE DB commits automatically).
7. A success message is shown and the four textboxes are cleared for the next entry.
8. catch displays the Oracle error text, for example ORA-00001 if the Book ID already exists.
9. Finally closes the connection whether the insert worked or failed.

Step 5: Button2 - Issue Book

```
Private Sub Button2_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button2.Click
    Dim bid As String = InputBox("Enter the Book ID to Issue:")
    If bid = "" Then Exit Sub

    Try
        cn.Open()
        Dim sql As String = "UPDATE book SET quan = quan - 1 WHERE bookid = " & bid
        Dim cmd As New OleDbCommand(sql, cn)
        Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

        If rowsAffected > 0 Then
            MsgBox("Book Issued Successfully!")
        Else
            MsgBox("No record found with ID: " & bid)
        End If
    Catch ex As Exception
        MsgBox("Error issuing book: " & ex.Message)
    Finally
        cn.Close()
    End Try
End Sub
```

1. `InputBox(...)` pops up a small window to type the Book ID, so the main textboxes need not be filled. If the user clicks Cancel or leaves it blank, `Exit Sub` stops.
2. The statement is `UPDATE book SET quan = quan - 1 WHERE bookid = 101`. It reduces the stock of that one book by 1.
3. `ExecuteNonQuery` returns how many rows were changed and it is stored in `rowsAffected`.
4. If `rowsAffected > 0` the book existed and was issued. If it is 0, no book has that ID, so 'No record found' is shown.

Step 6: Button3 - Return Book

```
Private Sub Button3_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button3.Click
    Dim bid As String = InputBox("Enter the Book ID to Return:")
    If bid = "" Then Exit Sub

    Try
        cn.Open()
        Dim sql As String = "UPDATE book SET quan = quan + 1 WHERE bookid = " & bid
        Dim cmd As New OleDbCommand(sql, cn)
        Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

        If rowsAffected > 0 Then
            MsgBox("Book Returned Successfully!")
        Else
            MsgBox("No record found with ID: " & bid)
        End If
    Catch ex As Exception
        MsgBox("Error returning book: " & ex.Message)
    Finally
        cn.Close()
    End Try
End Sub
```

Exactly the same as Issue, but the SQL is `quan = quan + 1`, so the stock goes up by 1.

Step 7: Button4 - View Report

```
Private Sub Button4_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button4.Click
    Try
        Dim sql As String = "SELECT * FROM book"
        Dim da As New OleDbDataAdapter(sql, cn)
        Dim dt As New DataTable()

        da.Fill(dt)
        DataGridView1.DataSource = dt
    Catch ex As Exception
        MsgBox("Error loading data: " & ex.Message)
    End Try
End Sub
End Class
```

1. `SELECT * FROM book` asks for all rows and all columns.
2. The `OleDbDataAdapter` takes the query and the connection. `New DataTable()` is an empty in-memory table.
3. `da.Fill(dt)` runs the query and copies the rows into the `DataTable`. There is no `cn.Open()` here because `Fill` opens and closes the connection itself.
4. `DataGridView1.DataSource = dt` displays the rows in the grid at the bottom of the form.
5. `End Class` closes the form class.

Sample run (output)

Action	Input	Result
Add Book	101, DBMS Concepts, Korth, 5	Message: Book Added Successfully!
Issue Book	101	Message: Book Issued Successfully! Quantity becomes 4
Return Book	101	Message: Book Returned Successfully! Quantity becomes 5
Issue Book	999 (does not exist)	Message: No record found with ID: 999
View Report	-	Grid shows BOOKID, BOOKNAME, AUTHOR, QUAN

Check in Oracle: `SQL> select * from book;`

Common errors in Program 8

Error	Reason and fix
ORA-00001 unique constraint violated	That Book ID is already in the table. Use a new ID.
ORA-00936 missing expression	A textbox was empty or the names of the textboxes were changed.
ORA-12154 / ORA-12541	Service name or listener problem. Start OracleXETNSListener, keep Data Source=XE.
ORA-01017 invalid username/password	Change User ID / Password in the connection string to your SYSTEM login.
MSDAORA provider not registered	Set Target CPU to x86 (common setup, note 1).

Viva questions (Program 8)

1. **What does ExecuteNonQuery do?** Runs INSERT/UPDATE/DELETE and returns the number of rows affected.
2. **Why is Finally used?** To close the connection always, even when an error happens.
3. **Why is rowsAffected checked?** To find out whether a book with that ID really exists.

- 4. What is the use of DataAdapter and DataTable?** The adapter runs SELECT and fills the DataTable, which the DataGridView displays.
- 5. What is MSDAORA?** The Microsoft OLE DB provider for Oracle.

Program 9: Student Mark Processing (st table)

Aim: To calculate total marks, result (PASS/FAIL) and percentage of a student from four subject marks, store the record in Oracle 10g XE, delete a record by Student ID and view all records.

Step 1: Create the table in Oracle

```
CREATE TABLE st (  
    stid    NUMBER PRIMARY KEY,  
    stname  VARCHAR2(50),  
    os      NUMBER,  
    rdbms   NUMBER,  
    cn      NUMBER,  
    se      NUMBER,  
    total   NUMBER,  
    result  VARCHAR2(10),  
    percent NUMBER  
);
```

Column	Meaning
stid	Student ID, primary key (unique).
stname	Student name, up to 50 characters.
os, rdbms, cn, se	Marks in the four subjects: Operating Systems, RDBMS, Computer Networks, Software Engineering.
total	Sum of the four marks.
result	PASS or FAIL (text, up to 10 characters).
percent	Percentage = total / 4.

The column order here is the order used in the INSERT statement in the code, so do not change it.

Step 2: Design the form (use the default names)

Control name	Used for	Filled by
TextBox1	Student ID	User
TextBox2	Student Name	User
TextBox3	OS mark	User
TextBox4	RDBMS mark	User
TextBox5	CN mark	User
TextBox6	SE mark	User
TextBox7	Total marks	Calculate button
TextBox8	Result (PASS/FAIL)	Calculate button
TextBox9	Percentage	Calculate button
Button1 / 2 / 3 / 4	Calculate / Insert / Delete / View	-
DataGridView1	Shows the records (bottom of the form)	View button

Use the default names while dragging the controls in this order. If the names are mixed up, the wrong values go into the wrong columns and Oracle may give ORA-00936.

Step 3: Code, connection

```
Imports System.Data.OleDb

Public Class Form1
    Dim cn As New OleDbConnection("Provider=MSDA0RA;Data Source=XE;User ID=system;Password=system;")
```

Step 4: Button1 - Calculate

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    TextBox7.Text = (Val(TextBox3.Text) + Val(TextBox4.Text) + Val(TextBox5.Text) +
        Val(TextBox6.Text)).ToString()

    If Val(TextBox3.Text) <= 30 Or Val(TextBox4.Text) <= 30 Or Val(TextBox5.Text) <= 30 Or
        Val(TextBox6.Text) <= 30 Then
        TextBox8.Text = "FAIL"
    Else
        TextBox8.Text = "PASS"
    End If

    TextBox9.Text = (Val(TextBox7.Text) / 4).ToString()
End Sub
```

1. `Val(TextBox3.Text)` converts the text in the box into a number. A blank box becomes 0, so there is no error.
2. **Total:** the four marks are added and `.ToString()` converts the answer back to text so it can be shown in `TextBox7`.
3. **Result:** if any one subject is 30 or below (`<= 30`), the result is FAIL. Only when all four are above 30 is it PASS. The word `or` is what makes one failed subject enough. Change 30 if your college uses a different pass mark.
4. **Percentage:** total divided by 4, because there are four subjects of 100 marks each.

Step 5: Button2 - Insert

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    Button1.PerformClick() ' auto-calculate before insert
    Try
        cn.Open()
        Dim sql As String = "INSERT INTO st VALUES (" & TextBox1.Text & ", '" & TextBox2.Text & "', " &
            TextBox3.Text & ", " & TextBox4.Text & ", " & TextBox5.Text & ", " & TextBox6.Text & ", " &
            & TextBox7.Text & ", '" & TextBox8.Text & "', " & TextBox9.Text & ")"
        Dim cmd As New OleDbCommand(sql, cn)
        cmd.ExecuteNonQuery()

        MsgBox("Successfully Inserted")

        TextBox1.Clear(): TextBox2.Clear(): TextBox3.Clear()
        TextBox4.Clear(): TextBox5.Clear(): TextBox6.Clear()
        TextBox7.Clear(): TextBox8.Clear(): TextBox9.Clear()
    Catch ex As Exception
        MsgBox(ex.Message)
    Finally
        cn.Close()
    End Try
End Sub
```

1. `Button1.PerformClick()` clicks the Calculate button by code, so Total, Result and Percentage are always filled before saving. The user does not have to press Calculate first.
2. `cn.Open()` opens the Oracle connection.
3. The INSERT string takes the nine textboxes in the same order as the table columns:

Textb ox	TextBox 1	TextBox 2	TextBox 3	TextBox 4	TextBox 5	TextBox 6	TextBox 7	TextBox 8	TextBox 9
Column	stid	sname	os	rdbms	cn	se	total	result	percent
Quotes?	no	yes	no	no	no	no	no	yes	no

1. Only the text columns sname and result are placed inside single quotes. For example: INSERT INTO st VALUES (1, 'Arun', 60, 70, 55, 80, 265, 'PASS', 66.25)
2. ExecuteNonQuery() saves the row. Then a message is shown.
3. The line TextBox1.Clear(): TextBox2.Clear(): ... uses a colon : to write several statements on one line. It clears all nine boxes.
4. catch shows any Oracle error (duplicate Student ID, empty box) and Finally closes the connection.

Step 6: Button3 - Delete

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
    Dim studentID As String = InputBox("Enter the Student ID to delete:")
    If studentID = "" Then Exit Sub

    Try
        cn.Open()
        Dim cmd As New OleDbCommand("DELETE FROM st WHERE stid = " & studentID, cn)
        Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

        If rowsAffected > 0 Then
            MsgBox("Record with ID " & studentID & " deleted successfully!")
        Else
            MsgBox("No record found with ID: " & studentID)
        End If
    Catch ex As Exception
        MsgBox(ex.Message)
    Finally
        cn.Close()
    End Try
End Sub
```

1. The Student ID is taken with an InputBox. Blank or Cancel stops the procedure.
2. The SQL is DELETE FROM st WHERE stid = 1. The WHERE part is important, without it every row would be deleted.
3. If rowsAffected > 0 the record was deleted, otherwise 'No record found' is shown.

Step 7: Button4 - View

```
Private Sub Button4_Click(sender As Object, e As EventArgs) Handles Button4.Click
    Try
        Dim da As New OleDbDataAdapter("SELECT * FROM st", cn)
        Dim dt As New DataTable()
        da.Fill(dt)
        DataGridView1.DataSource = dt
    Catch ex As Exception
        MsgBox(ex.Message)
    End Try
End Sub
End Class
```

Reads all rows of st into a DataTable and shows them in DataGridView1 (same method as Program 8).

Sample run (output)

Input (ID, name, OS, RDBMS, CN, SE)	Total	Result	Percentage
1, Arun, 60, 70, 55, 80	265	PASS	66.25
2, Divya, 25, 70, 55, 80	230	FAIL (OS is 25)	57.5

Delete: enter 2, the message 'Record with ID 2 deleted successfully!' appears. Check in Oracle: SQL> select * from st;

Common errors in Program 9

Error	Reason and fix
ORA-00936 missing expression	A textbox was empty (ID, name or a mark) or the textbox names are not the default ones.
ORA-00001 unique constraint	The Student ID already exists.
ORA-01722 invalid number	A mark or the ID contains letters.
ORA-00984 column not allowed here	The name was typed without quotes in the SQL. Keep the single quotes around sname and result.

Viva questions (Program 9)

1. **What is Val()?** Converts a text value to a number. Blank text gives 0.
2. **Why is PerformClick used in Insert?** To calculate total, result and percentage automatically before saving.
3. **Why are some values in single quotes in the SQL?** Character columns need quotes. Number columns do not.
4. **Why is WHERE used in DELETE?** To delete only the chosen student.
5. **How does the program decide FAIL?** If any one subject is 30 or below.

Program 10: Employee Payroll (emp table)

Aim: To calculate allowance, deduction, gross salary and net salary of an employee, store the employee record in Oracle 10g XE, delete a record by Employee ID, view all records and exit the application.

Step 1: Create the table in Oracle

```
CREATE TABLE emp (  
    empid    NUMBER PRIMARY KEY,  
    empname  VARCHAR2(50),  
    salary   NUMBER,  
    gross    NUMBER,  
    net      NUMBER  
);
```

Column	Meaning
empid	Employee ID, primary key (unique).
empname	Employee name, up to 50 characters.
salary	Basic salary typed by the user.
gross	Basic salary + allowance.
net	Gross salary - deduction.

Allowance and deduction are shown on the form but are **not stored** in the table.

Step 2: Design the form (default names)

Control name	Used for	Filled by
TextBox1	Employee ID	User
TextBox2	Employee Name	User
TextBox3	Basic Salary	User
TextBox4	Allowance (25% of basic)	Calculate button
TextBox5	Deduction (12% of basic)	Calculate button
TextBox6	Gross Salary	Calculate button
TextBox7	Net Salary	Calculate button
Button1 / 2 / 3 / 4 / 5	Calculate / Insert / Delete / View / Exit	-
DataGridView1	Shows the records (bottom of the form)	View button

Step 3: Code, connection

```
Imports System.Data.OleDb  
  
Public Class Form1  
    Dim cn As New OleDbConnection("Provider=MSDAORA;Data Source=XE;User ID=system;Password=system;")
```

Step 4: Button1 - Calculate

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click  
    TextBox4.Text = (Val(TextBox3.Text) * 0.25).ToString() ' Allowance  
    TextBox5.Text = (Val(TextBox3.Text) * 0.12).ToString() ' Deduction  
    TextBox6.Text = (Val(TextBox3.Text) + Val(TextBox4.Text)).ToString() ' Gross  
    TextBox7.Text = (Val(TextBox6.Text) - Val(TextBox5.Text)).ToString() ' Net  
End Sub
```

Line	Formula	Example (basic 10000)
Allowance (TextBox4)	basic x 0.25 (25%)	2500
Deduction (TextBox5)	basic x 0.12 (12%)	1200
Gross (TextBox6)	basic + allowance	12500
Net (TextBox7)	gross - deduction	11300

The order of the four lines matters. Gross uses the allowance that was just placed in TextBox4, and Net uses the gross in TextBox6 and deduction in TextBox5. So allowance and deduction must be calculated first. The text after the apostrophe ' is only a comment.

Step 5: Button2 - Insert

```
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    Button1.PerformClick() ' auto-calculate before insert
    Try
        cn.Open()
        Dim sql As String = "INSERT INTO emp VALUES (" & TextBox1.Text & ", '" & TextBox2.Text & "', " &
            TextBox3.Text & ", " & TextBox6.Text & ", " & TextBox7.Text & ")"
        Dim cmd As New OleDbCommand(sql, cn)
        cmd.ExecuteNonQuery()

        MsgBox("Successfully Inserted")

        TextBox1.Clear(): TextBox2.Clear(): TextBox3.Clear()
        TextBox4.Clear(): TextBox5.Clear(): TextBox6.Clear()
        TextBox7.Clear()
    Catch ex As Exception
        MsgBox(ex.Message)
    Finally
        cn.Close()
    End Try
End Sub
```

1. Button1.PerformClick() calculates all four values first.
2. The INSERT uses only five textboxes because the table has five columns:

empid	empname	salary	gross	net
TextBox1	TextBox2 (in quotes)	TextBox3	TextBox6	TextBox7

1. TextBox4 (allowance) and TextBox5 (deduction) are skipped on purpose. Only empname is a text value, so only it has single quotes. Example string: INSERT INTO emp VALUES (1, 'Ravi', 10000, 12500, 11300)
2. After ExecuteNonQuery() the message is shown and all seven boxes are cleared. Catch and Finally work as in the other programs.

Step 6: Button3 - Delete

```
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
    Dim eid As String = InputBox("Enter the EMPLOYEE ID to delete:")
    If eid = "" Then Exit Sub

    Try
        cn.Open()
        Dim cmd As New OleDbCommand("DELETE FROM emp WHERE empid = " & eid, cn)
        Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

        If rowsAffected > 0 Then
            MsgBox("Record with ID " & eid & " deleted successfully!")
        Else
            MsgBox("No record found with ID: " & eid)
        End If
    Catch ex As Exception
        MsgBox(ex.Message)
    Finally
        cn.Close()
    End Try
End Sub
```

The Employee ID is read with an InputBox and the statement DELETE FROM emp WHERE empid = 1 removes that row. rowsAffected tells whether the employee was found.

Step 7: Button4 - View

```
Private Sub Button4_Click(sender As Object, e As EventArgs) Handles Button4.Click
    Try
        Dim da As New OleDbDataAdapter("SELECT * FROM emp", cn)
        Dim dt As New DataTable()
        da.Fill(dt)
        DataGridView1.DataSource = dt
    Catch ex As Exception
        MsgBox(ex.Message)
    End Try
End Sub
```

Loads all employees into a DataTable and shows them in DataGridView1.

Step 8: Button5 - Exit

```
Private Sub Button5_Click(sender As Object, e As EventArgs) Handles Button5.Click
    Application.Exit()
End Sub
End Class
```

Application.Exit() closes the whole application. End Class closes the form class.

Sample run (output)

Input (ID, name, basic)	Allowance	Deduction	Gross	Net
1, Ravi, 10000	2500	1200	12500	11300
2, Meena, 20000	5000	2400	25000	22600

Check in Oracle: SQL> select * from emp; (shows empid, empname, salary, gross, net).

Common errors in Program 10

Error	Reason and fix
ORA-00936 missing expression	ID, name or basic salary was left empty, or textbox names were changed.
ORA-00001 unique constraint	The Employee ID already exists.
ORA-01722 invalid number	Salary contains letters.
Net or gross shows 0	Calculate lines were reordered, or basic salary is blank.

Viva questions (Program 10)

- 1. How are gross and net salary found?** Gross = basic + 25% allowance. Net = gross - 12% deduction.
- 2. Why are allowance and deduction not inserted?** The emp table has no columns for them. They are only calculated for display.
- 3. What does Application.Exit do?** Closes the application.
- 4. Why is the connection closed in Finally?** So that it is released even if an error occurs.
- 5. Which provider connects VB to Oracle here?** MSDAORA through the OleDb classes.